



In collaboration with MIT Club of Northern California and MIT Club of Germany

CHALLENGE

06

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Genome Firewall: An AI Defense System Against Superbugs

Use artificial intelligence (AI) to predict which antibiotics may fail from a bacterial genome — before standard lab results arrive

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Goals and Motivation

The Next Global Health Crisis Is Already Here

Antibiotics transformed modern medicine. Today, that foundation is beginning to crack. Antibiotic-resistant infections are associated with more than 4.7 million deaths each year, and over one million people die directly because existing antibiotics no longer work. If this trend continues, routine surgery, cancer treatment, organ transplants, and even common infections could become far more dangerous.

The challenge is not only that antibiotics can fail. It is that healthcare teams often do not know which antibiotic will work quickly enough. Standard laboratory testing usually takes one to three days. During that window, doctors must make their best-informed guess. Every ineffective treatment can cost a patient critical time and give resistant bacteria another opportunity to survive and spread.

Two Futures

Future 1 - We Get the AI Revolution Right

Imagine every hospital can analyze a bacterial genome in minutes. Healthcare professionals receive an AI-generated prediction of which antibiotics are most likely to work, together with a clear explanation, confidence score, and an honest no-call when the evidence is uncertain. Patients receive effective treatment sooner, unnecessary broad antibiotic use falls, and resistant outbreaks can be detected earlier. AI becomes a defensive layer for global biosecurity.

Future 2 - We Fail to Act

Now imagine nothing changes. Hospitals continue treating patients with incomplete information while resistant bacteria spread faster than laboratory results arrive. More infections become difficult - or impossible - to treat. Procedures that depend on reliable antibiotics become riskier, and the world begins to lose one of the greatest medical advances of the last century.

The Opportunity

Much of the answer is already written in a bacterium's deoxyribonucleic acid (DNA). Once its genome has been sequenced and reconstructed, AI can identify patterns that predict which antibiotics are likely to work - potentially days before standard laboratory testing is complete. Your challenge is to build a trustworthy AI defense system that turns a bacterial genome into an earlier antibiotic-response prediction, helping healthcare teams act faster and public-health teams track resistance sooner.

This challenge is strictly defensive. The system must never design, modify, or suggest changes to an organism. Its purpose is to help slow antibiotic resistance, protect patients, and strengthen global biosecurity.

Current Challenges

STANDARD LAB TESTING TAKES DAYS	Standard laboratory testing takes one to three days. Until the result arrives, treatment is based on the best available guess - and in a serious infection, every hour without an effective drug matters.
BEST-GUESS TREATMENT CAN DRIVE RESISTANCE	Using an antibiotic that targets many bacteria while waiting can save lives, but unnecessary use also increases resistance. Faster, targeted choices protect patients and preserve the drugs that still work.
GENOME DATA IS NOT YET TRUSTED DECISION SUPPORT	Sequencing is increasingly fast and affordable, and a bacterial genome contains many clues about resistance. The missing step is a trusted system that turns a reconstructed genome into a clear, well-calibrated prediction for each antibiotic.

Your Challenge

Build GENOME FIREWALL - a research prototype that takes a reconstructed, quality-checked genome from ONE supported bacterial species and predicts which antibiotics are likely to work, likely to fail, or remain uncertain. Each result must include a well-calibrated confidence score, an explicit no-call option, and the genes or DNA changes supporting the prediction. The prediction should also account for the presence of the drug's molecular target, so the system does not report "likely to work" based solely on the absence of resistance markers. This is protective decision support for faster, more targeted antibiotic use and earlier resistance tracking - never organism design or modification.

In scope vs out of scope - read this first

IN SCOPE: a quality-checked FASTA file (the standard plain-text format for DNA and protein sequences) containing one reconstructed bacterial genome -> for each antibiotic, likely to fail / likely to work / no-call, with a confidence score and supporting genes or DNA changes. **OUT OF SCOPE:** collecting samples, reading DNA directly from blood, identifying the bacterial species, reconstructing the genome, or separating multiple bacteria in one sample. Your system starts only after bacterial isolation, sequencing and genome reconstruction are complete.

Build these modules:

01 The Genome Reader - From DNA to AI Features

Turn a reconstructed bacterial genome into features an AI model can use. The gold standard is AMRFinderPlus (developed by the National Center for Biotechnology Information (NCBI); public-domain, unrestricted); it identifies antimicrobial-resistance (AMR) genes and resistance-associated mutations using protein annotations and/or assembled nucleotide sequence. Can you build an AI model that replaces or improves upon AMRFinderPlus? Whether you choose to rely on AMRFinderPlus or build an alternative, the next step is to develop a model that takes the presence/absence of known AMR genes or mutations as input.

Required: A documented, repeatable path from an assembled FASTA file to model features on the provided fixed dataset, using AMRFinderPlus as the default annotation tool, and a specification for the output format.

02 The Predictor - Will Each Antibiotic Work?

Compile a database of available antimicrobial drugs and their properties. For each set of features produced by Module 01 from a bacterial species FASTA, generate predictions for each drug (likely to fail / likely to work / no-call). Ensure antibiotic compatibility by applying a deterministic gate: the presence or absence of the drug's molecular target. The training stage will need a de-duplication step based on sequence homology, so the model maximizes diversity in the training set by recognizing sequences seen previously, rather than memorizing closely related genomes.

Required: Predictions for all antibiotics on each species, evaluated after a de-duplication clustering step. Identical or near-identical genomes should not appear in both training and testing - the sequence-homology threshold used for de-duplication is left to each team to tune and justify.

03 The Decision Report - Explain Confidence and Know When Not to Predict

Present every result as a clear antibiotic-response report: the drug, likely to fail / likely to work / no-call, a calibrated confidence score, and the type of evidence behind it - (i) a known resistance gene or DNA change was detected, (ii) the model found only a statistical association, or (iii) no known resistance signal was found. Forcing every sample into a yes/no answer creates false confidence; returning no-call for weak or conflicting evidence is a strength. Build a small app that tells users every result must be confirmed by standard lab testing.

Required: A working Streamlit or Gradio demo that returns likely to fail / likely to work / no-call for each drug, with calibrated confidence and an evidence category, plus a mandatory "confirm with standard lab testing" message.

The Responsibility Requirement

This is a biosecurity challenge with a strictly protective goal. Your prototype must address the following:

- **Defensive by construction:** the tool predicts and explains resistance that already exists to support treatment choices and public-health tracking. It must never generate, design, or suggest changes to an organism.
- **Honest generalization:** report performance using the provided split by genetically related groups, ideally including groups the model has not seen before. State clearly which bacterial species and antibiotics the system does and does not cover.
- **Calibrated confidence and a no-call option:** a confident but wrong result could point a care team toward the wrong drug. Show that confidence scores match real performance, and return no-call when the evidence is weak, conflicting, or unlike the training data.
- **Honest explanations:** clearly separate a known resistance gene or DNA change from a feature that is only statistically associated with resistance. A feature-importance score or SHAP (SHapley Additive exPlanations) value does not automatically prove a biological cause.
- **Human oversight:** the antibiotic-response report is decision support that must be confirmed by a trained healthcare or laboratory professional. It must never make a treatment decision on its own.

Show these on the held-out data in your demo and explain how you addressed each.

OpenAI Tools & Credits

This challenge is powered by OpenAI. We recommend exploring OpenAI's models for this challenge. Hack-Nation is providing \$50 in free OpenAI application programming interface (API) credits per team, available on a first-come, first-served basis.

- Be creative in exploring multimodal approaches and combining data inputs across domains - combine OpenAI's multimodal capabilities and image generation (gpt-image-2) with other modality models.
- Strong entries show multimodal integration, user value, technical quality, creativity, responsible design, and demo quality.

Data Sources and Hints

Build your system using real, openly available bacterial genome data. Organizers will provide a fixed challenge dataset (see Appendix) so teams can focus on modeling rather than spending the event collecting and cleaning large raw archives.

GENOME + RESISTANCE DATA

BV-BRC (ex-PATRIC)	BV-BRC (Bacterial and Viral Bioinformatics Resource Center; formerly PATRIC, the Pathosystems Resource Integration Center) is our primary source: 15,000+ bacterial genomes linked to laboratory results showing whether antibiotics worked, including measurements of how much drug was needed to stop growth. Public data are freely available; bulk download uses anonymous FTPS (File Transfer Protocol Secure). Use the organizer-pinned, laboratory-measured test results - NOT general phenotype fields, which may contain model-generated predictions. bv-brc.org
AMRFinderPlus (default)	NCBI tool and database that detects known resistance genes and small DNA changes. It is described as public-domain and unrestricted and is the recommended default annotation tool for this event. github.com/ncbi/amr
ResFinder	ResFinder identifies acquired genes and/or finds chromosomal mutations mediating antimicrobial resistance in total or partial DNA sequence of bacteria. genepi.food.dtu.dk/resfinder
cAMRah	cAMRah is a curated workflow designed to predict AMR genes in microbial genomes; it integrates and runs six AMR-finding tools and databases: AMRFinderPlus, ResFinder, RGI (Resistance Gene Identifier) with the CARD (Comprehensive Antibiotic Resistance Database) database, Abricate with the NCBI database, Abricate with the ARG-ANNOT (Antibiotic

	Resistance Gene-ANNOTation) database, and the BV-BRC AMR detection tool. pmc.ncbi.nlm.nih.gov/articles/PMC12910510/
XTree	k-mer-based aligner designed for memory-efficient (and virtual/out-of-core memory) parallel alignment of long or short sequencing reads to millions of genomes. github.com/two-frontiers-project/2FP-XTree
Kaggle mirrors (optional)	Suitable for tutorials only. Use a mirror only when the dataset license, version, and original source are clearly documented. A Kaggle copy alone is not a verified open benchmark.
MODELS & COMPUTE	
Baseline (recommended)	Use one regularized logistic-regression model per antibiotic with features from AMRFinderPlus, such as known resistance genes and DNA changes. It runs on a central processing unit (CPU) and is fast, easy to calibrate, and easy to explain - a dependable core approach.
Deep-learning stretch (optional)	Genomic language models such as HyenaDNA (context lengths from 32k up to 1 million bases in the largest checkpoint) and DNABERT-2 (117M parameters) still cannot read an entire bacterial genome in one pass. A realistic advanced option is to create embeddings for selected, annotated DNA regions or genome chunks, combine them into one representation per bacterial sample, and compare the result with the baseline. Alternatives include translating the genome to a proteome and running a protein-sequence model on the result, or hashing the genome sequence into k-mers to collapse repetitive non-coding regions.
Compute	Baseline: CPU. Advanced deep-learning option: one graphics processing unit (GPU) using selected DNA regions or chunks. Free Colab does not guarantee GPU access, so organizers should keep deep learning optional and provide limited backup compute if they expect deep-learning submissions.

Success Criteria

Judge submissions on safe, honest performance - not a single headline accuracy number from an unbalanced dataset. Report and be evaluated on:

- **Balanced accuracy**, plus recall for resistant cases (the antibiotic is likely to fail) and susceptible cases (the antibiotic is likely to work), reported separately.
- **F1 score** (the harmonic mean of precision and recall), AUROC (area under the receiver operating characteristic curve), and PR-AUC (precision-recall area under the curve) per drug (PR-AUC matters under class imbalance).
- **Confidence quality**: Brier score and a reliability plot; also report how often the system returns no-call and how accurate the remaining predictions are.
- **Generalization**: performance broken down by genetically related bacterial groups, using the organizer's hidden test set with groups not seen during training.

Note: published baseline models perform strongly for some well-documented bacteria and antibiotics, but results depend on label quality, class balance, genetic similarity between samples, and how the data is split. There is no guaranteed target score - report results on the fixed held-out test split.

What Makes a Strong Submission

Strong submissions...	Weak submissions...
— Use the grouped split and report results by genetically related group. Show that the model learns resistance signals rather than memorizing nearly identical bacteria.	— Split rows randomly, allow nearly identical genomes into both training and testing, and report an inflated score.

Strong submissions...

- Do one bacterial species and a few antibiotics well, with calibrated confidence and a no-call option.
- Explain results honestly. Separate known resistance genes or DNA changes from features that are only statistically associated.
- Keep the system strictly defensive and say so. Frame it as support for targeted antibiotic use and public-health tracking.

Weak submissions...

- Claim coverage for every disease-causing bacterium and every antibiotic, force every sample into a yes/no answer, and hide uncertainty.
- Present a raw SHAP or feature-importance score as if it proved a biological cause of resistance.
- Drift toward designing, changing, strengthening, or optimizing organisms.

Why This Matters

Antibiotic resistance is a slow-moving global crisis already underway. More than one million deaths each year are directly attributed to it, and the World Health Organization (WHO) identifies it as a major threat to human health. Two actions matter most: get the right antibiotic to the right patient sooner, and reduce unnecessary use of broad treatments while the lab catches up. A genome-to-antibiotic-response tool supports both by providing an earlier, evidence-based prediction and helping public-health teams see where resistance is spreading. The data is open, a reliable baseline can be built during a weekend, and the goal is clearly defensive.

Appendix — Organizer Notes, Contact & Resources

Research prototype. Predictions based on historical bacterial genome data do not prove that the system is safe, accurate enough, approved, or suitable for real healthcare decisions. Every antibiotic-response report must be confirmed with standard laboratory testing.

For organizers - provide a fixed challenge dataset (this is the difference between a productive event and 24 hours spent cleaning data):

- 1,000-3,000 reconstructed genomes for ONE bacterial species; 3-5 predefined antibiotics; only outcomes measured in a laboratory.
- Standardized antibiotic names; a documented rule for likely to fail / likely to work / uncertain; one final label for each genome-antibiotic pair; genome quality flags; source and accession records; and a license and attribution file.
- Group genomes by genetic similarity, with fixed training, confidence-calibration, and HIDDEN test sets; include file checksums and a repeatable download script.
- Ideally provide precomputed AMRFinderPlus results and, optionally, a precomputed sparse k-mer feature matrix; include a hidden evaluation set scored with the metrics above and run the full reference pipeline before the event under the same computing limits participants will face.

Scope & safety note: Limited to predicting and explaining resistance that already exists using reconstructed, openly available bacterial genomes. The tool may support treatment choices and public-health tracking, but it excludes all sample-to-genome processing and any design, synthesis, or enhancement of organisms.